THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · EASY LEVEL CHAPTER TEST SET 3

Chapter 10: **The Human Eye and the Colourful World** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The bluish colour of the sky is due to 1

(a) reflection **(b)** scattering **(c)** refraction only **(d)** dispersion

2. In dim light the pupil 1

(a) becomes smaller **(b)** becomes larger **(c)** disappears **(d)** closes

3. The unit of power is 1

(a) ohm **(b)** watt **(c)** metre **(d)** dioptre

4. The reddish appearance of the Sun at sunrise is due to 1

(a) scattering **(b)** diffraction **(c)** dispersion **(d)** interference

5. In bright light the pupil 1

(a) becomes smaller **(b)** becomes a slit only

(c) stays the same **(d)** becomes larger

6. Bifocal lenses are used for 1

(a) astigmatism only **(b)** colour blindness **(c)** presbyopia **(d)** cataract

7. Stars twinkle because of 1

(a) clouds **(b)** the moon

(c) their own pulsing **(d)** atmospheric refraction

8. The eye lens is a 1

(a) convex lens **(b)** concave lens **(c)** plane mirror **(d)** prism

9. Good vision is helped by 1

(a) food rich in sugar **(b)** staying in dim light

(c) rubbing eyes **(d)** food rich in vitamin A

10. A rainbow is seen after 1

(a) rain, with the Sun behind us **(b)** a total eclipse

(c) midnight **(d)** a dry summer

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does the sky look blue? 2

12. What is the far point of a normal eye? 2

13. What is a spectrum? 2

14. What causes twinkling of stars? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A person is advised to use a lens of power +2 D. Find the focal length and the defect. 3

16. Why do we get advance sunrise? 3

17. What will happen if a prism is placed to receive a narrow beam of red light? 3

OR

A student cannot see the blackboard clearly from the last bench but can read a book easily. Name the defect and the lens. 3

18. What is a rainbow? How is it formed? 3

OR

Why do the planets not twinkle? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A person has a far point at 50 cm. Find the power of the lens needed to see distant objects. 5

OR

Explain why stars twinkle but planets do not. 5

20. Explain dispersion of white light through a glass prism and name the colours of the spectrum. 5

OR

What are bifocal lenses? Which defect do they correct and why? 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND